

How Technology Advance Is Calculated

Facilitator reference. Everything here is taken directly from `public/js/engine.js` → `_applyRoadmapAdvance()`, and every number below was produced by running the real engine. If someone in the room asks "*so how is the score actually calculated?*", this is the answer.

The last section, Hints to give players, is written to be read *aloud* — use it during the game without giving the whole formula away.

1. The one-line answer

Each round, your roadmap gains **R&D Capacity ÷ 5** — but only if none of your eight stats is in danger. One stat in danger freezes it. Two or more push it backwards.

That is the honest short version and it is usually enough.

2. The full equation

Run once per team, at the end of every round, **after** all event effects, card effects and interaction effects have been applied.

Step 0 — Definitions

The eight stats: treasury, energy, politicalSupport, publicWelfare, rdCapacity, reputation, security, environment

$T(k)$ = danger threshold of stat k (see section 3)
 D = DANGER COUNT stats below their threshold
 N = NEAR-MISS COUNT stats below threshold + 5
(only counted when $D = 0$)
spread = highest stat - lowest stat, over all eight
 P = current roadmap progress, 0 to 100
 A = this round's advance (what we solve for)

Note: all eight stats have a threshold — **R&D Capacity included**. Letting R&D itself fall below 20 both freezes the roadmap *and* shrinks the base rate.

Step 1 — Base rate

```
A = rdCapacity / 5
```

Step 2 — The balance gate (this is the important one)

```
if D = 1           A = 0           status = BLOCKED
else if D >= 2     A = -(D - 1) * 5 status = REGRESSING
else if D = 0 and N > 0 A = A / 2   status = STRAINED
else               A stays as it is status = STABLE
```

Step 3 — Imbalance penalty

```
if spread > 45     A = A - 3
```

Applies in **every** status, including REGRESSING (making it worse). It is the one penalty that can bite while all eight stats are still technically safe.

Step 4 — Round, then enforce the freeze

```
A = round(A)           the only rounding; all above is exact
if status = BLOCKED    A = 0   (frozen, never negative)
```

This ordering matters: a blocked team with a huge spread still moves **0**, not -3.

Step 5 — Apply

```
P = clamp(P + A, 0, 100)
```

Step 6 — Development upkeep (only if A > 0)

```
treasury = treasury - round(A / 2.0)
energy   = energy   - round(A / 2.8)
```

Building is not free. This is usually what eventually drags a fast-moving team into the danger zone in step 2 — the mechanism is self-limiting by design.

The whole thing as one expression

For the common case ($D = 0$, no near-miss, $spread \leq 45$):

```
delta = round( rdCapacity / 5 )
cost = ( -round(delta / 2.0) treasury , -round(delta / 2.8) energy )
```

Full form:

```

      0                                if D = 1          (BLOCKED)
      -(D - 1) * 5                     if D >= 2         (REGRESSING)
A    = (rdCapacity / 5) / 2           if D = 0, N > 0     (STRAINED)
      rdCapacity / 5                  otherwise          (STABLE)

A    =  A - 3                          if spread > 45

delta =  round(A)
delta =  0                             if status is BLOCKED
```

3. Reference tables

Danger thresholds

Stat	In danger below	Warning shown
 Treasury	20	Fiscal crisis
 Energy / Compute	20	Power / compute shortage
 R&D Capacity	20	Research capacity collapse
 Political Support	25	Political instability
 Public Welfare	25	Social unrest
 International Reputation	25	Diplomatically isolated
 Security / Sovereignty	25	Sovereignty at risk
 Environment	25	Environmental backlash

"Near-miss" = within 5 points above the threshold (so 20–24 for a 20-threshold stat, 25–29 for a 25-threshold stat).

What each status does

Stats in danger	Badge	Effect on roadmap
0, all clear by 5+	 Stable	Full rate
0, but ≥ 1 within 5 of a threshold	 Strained	Halved
1	 Blocked	Frozen at 0
2	 Regressing	-5
3	 Regressing	-10
4	 Regressing	-15

Base rate by R&D Capacity (*verified output*)

R&D Capacity	Advance / round	Upkeep cost
30	+6	-3 treasury, -2 energy
40	+8	-4 treasury, -3 energy
50	+10	-5 treasury, -4 energy
60	+12	-6 treasury, -4 energy
70	+14	-7 treasury, -5 energy
80	+16	-8 treasury, -6 energy
90	+18	-9 treasury, -6 energy

Roadmap progress → stage and card difficulty

Progress	Stage shown	Policy cards drawn from
0–19%	1 / 5	early
20–39%	2 / 5	early
40–59%	3 / 5	mid
60–79%	4 / 5	late
80–100%	5 / 5	late

4. Worked examples

All figures below are actual engine output.

A — Healthy team, R&D 60, everything clear

$A = 60/5 = 12$, $D=0$, $N=0$, spread 6
→ +12 progress (40% → 52%)
→ upkeep: -6 treasury, -4 energy

B — Same team, but Public Welfare has slipped to 27 (*threshold 25, so within 5 → near-miss*)

$A = 12$ → halved → 6
→ +6 progress (40% → 46%)
→ upkeep: -3 treasury, -2 energy

C — Public Welfare falls to 20 (*now genuinely below threshold*)

$D = 1$ → BLOCKED
→ +0 progress (stays 40%)
→ no upkeep charged

D — Welfare 20 and Treasury 15

$D = 2$ → $A = -(2-1)*5 = -5$
→ -5 progress (40% → 35%)

E — Three stats down (Welfare 20, Treasury 15, Energy 15)

$D = 3$ → $A = -(3-1)*5 = -10$
→ -10 progress (40% → 30%)

F — Imbalance alone, nothing in danger (R&D 60, Treasury 40, Energy 86)

$A = 12$, $D = 0$, spread = $48 > 45$ → -3
→ +9 progress

G — Regressing *and* badly unbalanced ($D=2$, spread 80)

$A = -5$, then -3 for spread
→ -8 progress (40% → 32%)

5. Edge cases worth knowing

- **Blocked never goes negative.** A blocked team with a 75-point spread still moves exactly 0 — the freeze is applied *after* the spread penalty.
 - **Upkeep is charged on the advance, not on the progress actually gained.** A team at 96% that earns +12 is clamped to 100% (gaining only 4) but still pays the full -6 / -4 upkeep. Finishing the roadmap is expensive.
 - **No upkeep when frozen or regressing.** Upkeep only applies when $A > 0$.
 - **Rounding happens once**, at the very end of step 4 — not at each intermediate step. So R&D 61 strained with a spread penalty is $61/5 = 12.2 \rightarrow 6.1 \rightarrow 3.1 \rightarrow \text{round} \rightarrow 3$.
 - **Every stat is clamped to 0–100** and rounded whenever it changes.
 - **R&D Capacity is itself gate-able.** Below 20 it is a danger stat *and* the base rate is ≤ 4 . This is the one genuine death spiral in the game.
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6. How the final ranking uses this

```
balance = mean of all eight stats  
SCORE   = roadmapProgress * 0.5 + balance * 0.5
```

Example: roadmap 80%, all stats at 50 $\rightarrow 80*0.5 + 50*0.5 = 65$.

Half the score is what you built; half is the state you left the country in. A team that races to 90% on a wrecked country and a team that keeps a perfect country at 20% both score badly.

7. Hints to give players

Ordered from gentlest to most explicit. Give them out as the room needs them — the discovery is part of the exercise.

Round 1–2: nudges (no numbers)

- *"Progress comes from R&D Capacity — but only your country lets it."*
- *"Watch the badge on your panel. Green is not the only state it has."*

- *"Nothing on this board is free. Look at what your last decision cost you."*

Round 3–4: mechanics (still no formula)

- *"A single stat in the danger zone stops your roadmap completely. Not slows — stops."*
- *"Two stats down and you start losing work you already did."*
- *"Being just barely safe is not safe. Sitting near the line already halves your speed."*
- *"Your R&D Capacity is a stat like any other. It can collapse too — and if it does, you're stuck twice over."*

Round 5+: full transparency (if they ask, tell them)

- *"You gain $R\&D \div 5$ per round. At R&D 60 that's 12 points."*
- *"Every point of progress costs you about half a point of treasury and a third of a point of energy."*
- *"If your best and worst stat are more than 45 apart, you lose 3 more on top."*
- *"Blocked is 0. Regressing is minus 5 for the second danger stat, minus 5 again for each one after."*

Strategic hints (the useful ones)

- **Repair before you build.** A round spent fixing one stat is worth more than a round of blocked progress. Blocked rounds are the single biggest source of lost progress.
- **Aim for a 5-point buffer,** not a 0-point one. The near-miss rule means "just above the line" still costs you half your speed.
- **Keep your stats close together.** Two teams with the same average can differ by 3 progress a round purely on spread.
- **The faster you build, the harder upkeep bites.** High R&D drains treasury and energy fastest — the strongest team is usually the one that trips first.
- **Gamble cards can undo three good rounds in one bad roll.** Bet when you're behind, not when you're ahead.

The line that usually lands

"You can't sprint out of a collapsing country. Fix it first — then run."

8. Where this lives in the code

What	Where
The formula	<code>public/js/engine.js</code> → <code>_applyRoadmapAdvance()</code>
Thresholds	<code>public/js/engine.js</code> → <code>THRESHOLDS</code>
Danger / near-miss counting	<code>dangerCount()</code> , <code>balanceStatus()</code>
Stage bands	<code>STAGE_BANDS</code> , <code>getRoadmapStage()</code> , <code>stageBand()</code>
Final score	<code>_computeDebrief()</code>

If you change any constant in the engine, update this file — the two are meant to agree, and the worked examples above were generated from the code as it stands.